

Viking Sports — AI Deal Analyzer

April 2026
CONFIDENTIAL

Model Performance Report | Machine Learning Investment Signal for Sports Memorabilia

0.872

Test AUC (ROC)

77.4%

Test Accuracy

0.774

F1 Score

12,089

Training Rows

// WHAT IT DOES

- **XGBoost gradient-boosted decision tree** model that predicts whether a sports memorabilia deal is a **BUY** or **NOT BUY**
- **5 inputs:** Asset Type (12 categories), Hold Period (1-40 yrs), Purchase Price Tier (5 tiers), Acquisition Decade (1980s-2020s), Deal Status
- Returns **probability score** (0-100%) with **confidence level** (High / Medium / Low)
- Runs **entirely in-browser** with < 2 second inference — no server calls required

// TRAINING DATA

Source	Rows	Description
Original Deals	89	Real Viking Sports historical transactions
Synthetic Augmentation	12,000	Generated via conditional probability tables from original patterns
Combined Dataset	12,089	80/20 stratified train-test split (9,671 / 2,418)

Class balance: 6,088 NOT BUY vs 6,001 BUY (near-equal)

// MODEL SELECTION PROCESS

- Tested **2 algorithms:** XGBoost and LightGBM
- Tested **3 resampling strategies:** SMOTE, SMOTETomek, No resampling
- **100 Optuna Bayesian optimization trials** per combination (600 total trials)
- **5-fold stratified cross-validation** for robust, unbiased evaluation

WINNER: XGBoost + No Resampling — depth=3, 800 trees, lr=0.016, Optuna-tuned across 9 hyperparameters

// MODEL LEADERBOARD

Rank	Algorithm	Sampling	Test AUC	Accuracy	F1	Precision	Recall
1	XGBoost	None	0.872	77.4%	0.774	76.9%	77.8%
2	LightGBM	SMOTE	0.871	77.6%	0.776	77.0%	78.2%
3	XGBoost	SMOTE	0.871	77.4%	0.776	76.5%	78.8%
4	LightGBM	None	0.870	77.3%	0.773	76.8%	77.8%
5	LightGBM	SMOTETomek	0.865	77.4%	0.774	76.9%	77.9%
6	XGBoost	SMOTETomek	0.864	77.4%	0.774	77.0%	77.8%

Performance Deep Dive

Detailed metrics, feature analysis, and prediction examples

// KEY METRICS

Metric	Score
Test AUC (ROC)	0.872
Test Accuracy	77.4%
Test F1 Score	0.774
Test Precision	76.9%
Test Recall	77.8%
Cross-Val AUC	0.871 +/- 0.003
Cross-Val Accuracy	77.9% +/- 0.5%

Confusion Matrix

	Pred: NOT BUY	Pred: BUY
Actual: NOT BUY	937 (TN)	281 (FP)
Actual: BUY	266 (FN)	934 (TP)

// TOP FEATURE IMPORTANCES

Rank	Feature	Importance	
1	Acquisition Decade	10.3%	
2	Tickets & Passes (type)	9.0%	
3	Is Rookie Card	8.3%	
4	Hold Period Bucket	7.6%	
5	Price Tier	5.5%	

// SAMPLE PREDICTIONS

Demonstrating model discrimination across different deal profiles:

Asset Type	Hold	Decade	Price Tier	Probability	Signal
Rookie Cards	10 yr	2000s	Under \$50	94.1%	BUY
Memorabilia	20 yr	2000s	\$50-\$500	98.9%	BUY
Publications	30 yr	1990s	Under \$50	98.2%	BUY
Stickers	8 yr	2010s	\$50-\$500	78.9%	BUY
Tickets & Passes	5 yr	2010s	\$50-\$500	39.8%	NOT BUY
Rookie Cards	2 yr	2020s	\$500-\$5K	15.9%	NOT BUY
Cards (Non-Rookie)	1 yr	2020s	\$5K-\$20K	13.6%	NOT BUY
Game-Worn Jerseys	3 yr	2020s	\$5K-\$20K	1.4%	NOT BUY

// PATH TO HIGHER ACCURACY

Current ceiling (~78%) is driven by synthetic data limitations, not model architecture. Improvements to push past 85%+:

- More real data — Expand from 89 to 300-500+ real deals for genuine market signal
- CTGAN synthetic generation — Neural network learns real distributions vs. hand-coded rules
- Richer features — Player popularity, sport type, condition grade, market timing
- Ensemble stacking — XGBoost + LightGBM + Random Forest with meta-learner

// OPTIMIZED HYPERPARAMETERS

max_depth: 3 | n_estimators: 800 | learning_rate: 0.016 | subsample: 0.66 | colsample_bytree: 0.61 | min_child_weight: 7 | gamma: 0.47 | reg_alpha: 0.042 | reg_lambda: 0.007